12 Advanced search

This chapter covers

- Geo data types
- Searching locations and addresses with geoqueries
- Using `geo_shape` to search for 2D shapes
- Using `span` queries to work with low-level positional tokens
- Specialized queries such as percolators

Earlier chapters have covered searching data using term-level and full-text queries. We've also looked at advanced queries like `bool`, `boosting`, and others. To continue building on what we've discussed and advance the query landscape, this chapter introduces several kinds of specialized queries.

We begin by looking at searches aimed at geolocations. Common use cases involving *geoqueries* include searching nearby restaurants for a delivery order, finding directions to a friend's house, locating popular schools within a 10 km range, and so on. Elasticsearch has first-class support for satisfying such location-related searches. It also provides several geospatial queries: `geo_bounding_box`, `geo_distance`, and `geo_shape`.